

The first line sets the value of `car` to `'bmw'` using a single equal sign, as you've seen many times already. The next line checks whether the value of `car` is `'bmw'` by using a double equal sign (`==`). This *equality operator* returns `True` if the values on the left and right side of the operator match, and `False` if they don't match. The values in this example match, so Python returns `True`.

When the value of `car` is anything other than `'bmw'`, this test returns `False`:

```
>>> car = 'audi'
>>> car == 'bmw'
False
```

A single equal sign is really a statement; you might read the first line of code here as “Set the value of `car` equal to `'audi'`.” On the other hand, a double equal sign asks a question: “Is the value of `car` equal to `'bmw'`?” Most programming languages use equal signs in this way.

Ignoring Case When Checking for Equality

Testing for equality is case sensitive in Python. For example, two values with different capitalization are not considered equal:

```
>>> car = 'Audi'
>>> car == 'audi'
False
```

If case matters, this behavior is advantageous. But if case doesn't matter and instead you just want to test the value of a variable, you can convert the variable's value to lowercase before doing the comparison:

```
>>> car = 'Audi'
>>> car.lower() == 'audi'
True
```

This test will return `True` no matter how the value `'Audi'` is formatted because the test is now case insensitive. The `lower()` method doesn't change the value that was originally stored in `car`, so you can do this kind of comparison without affecting the original variable:

```
>>> car = 'Audi'
>>> car.lower() == 'audi'
True
>>> car
'Audi'
```

We first assign the capitalized string `'Audi'` to the variable `car`. Then, we convert the value of `car` to lowercase and compare the lowercase value to the string `'audi'`. The two strings match, so Python returns `True`. We can see that the value stored in `car` has not been affected by the `lower()` method.

Websites enforce certain rules for the data that users enter in a manner similar to this. For example, a site might use a conditional test like this to